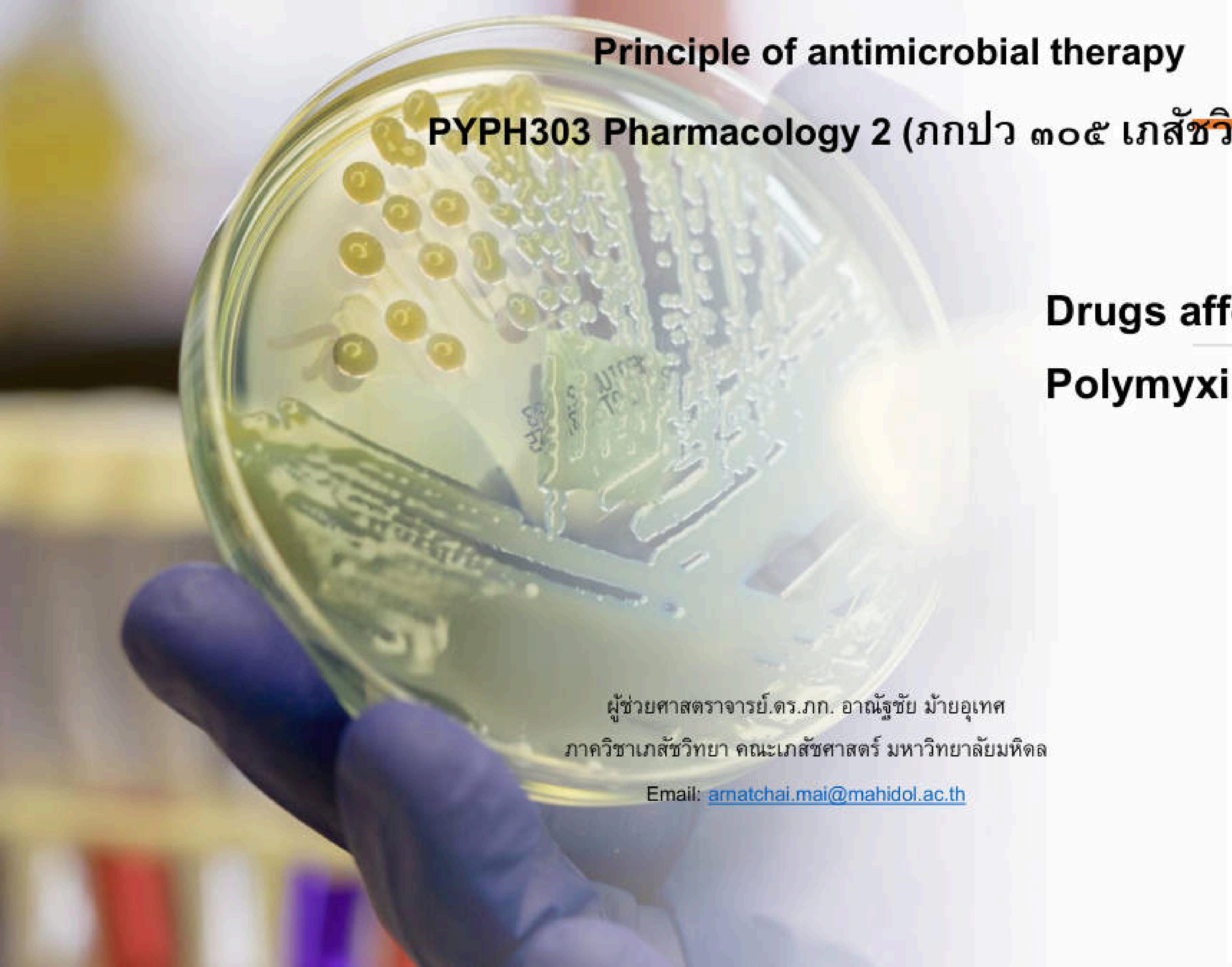

Principle of antimicrobial therapy

PYPH303 Pharmacology 2 (ภกปว ๓๐๕ เภสัชวิทยา 2)

Drugs affecting cell membrane:

Polymyxin, Daptomycin

ผู้ช่วยศาสตราจารย์.ดร.ภก. อาณัฐชัย ม้ายอุเทศ

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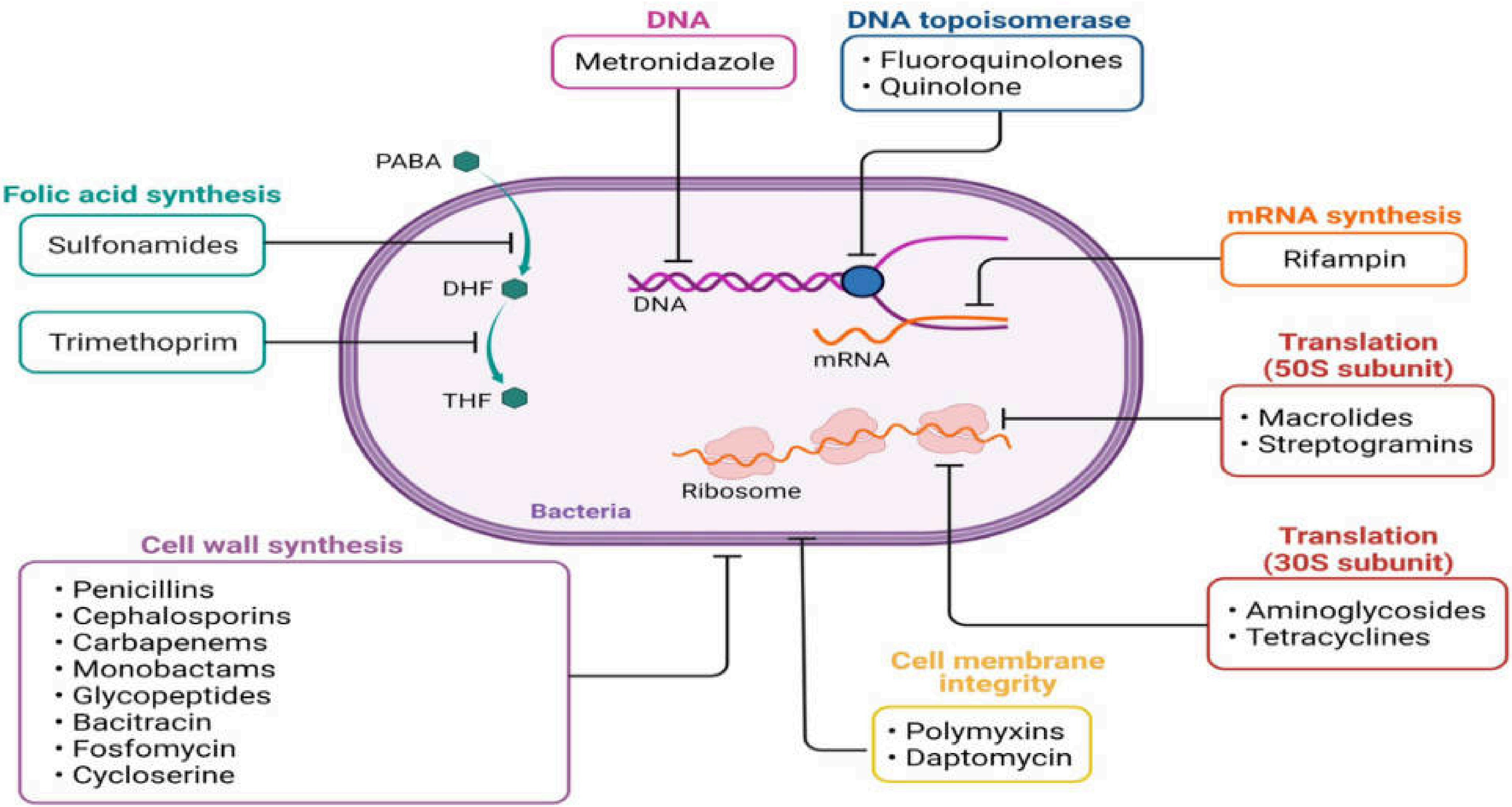
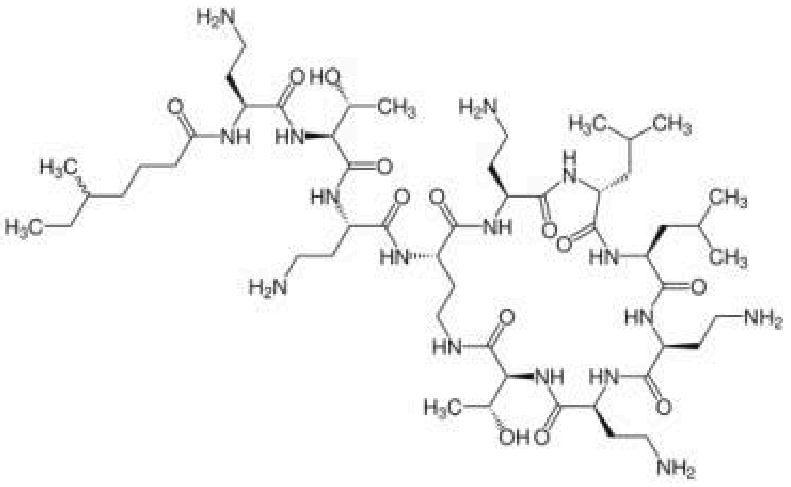
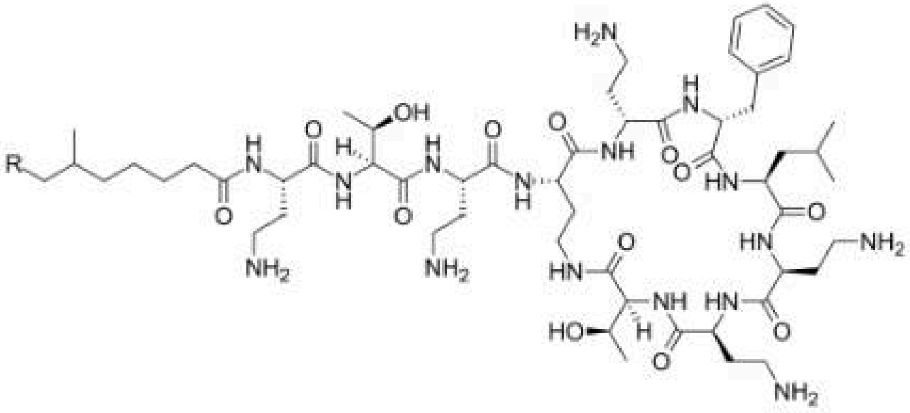

DNA
Metronidazole
DNA topoisomerase
• Fluoroquinolones
• Quinolone
PABA
Folic acid synthesis
Sulfonamides
DHF
Trimethoprim
THF
DNA
mRNA synthesis
Rifampin
mRNA
Translation
(50S subunit)
• Macrolides
• Streptogramins
Ribosome
Bacteria
Cell wall synthesis
• Penicillins
• Cephalosporins
• Carbapenems
• Monobactams
• Glycopeptides
• Bacitracin
• Fosfomycin
• Cycloserine
Translation
(30S subunit)
• Aminoglycosides
• Tetracyclines
Cell membrane
integrity
• Polymyxins
• Daptomycin



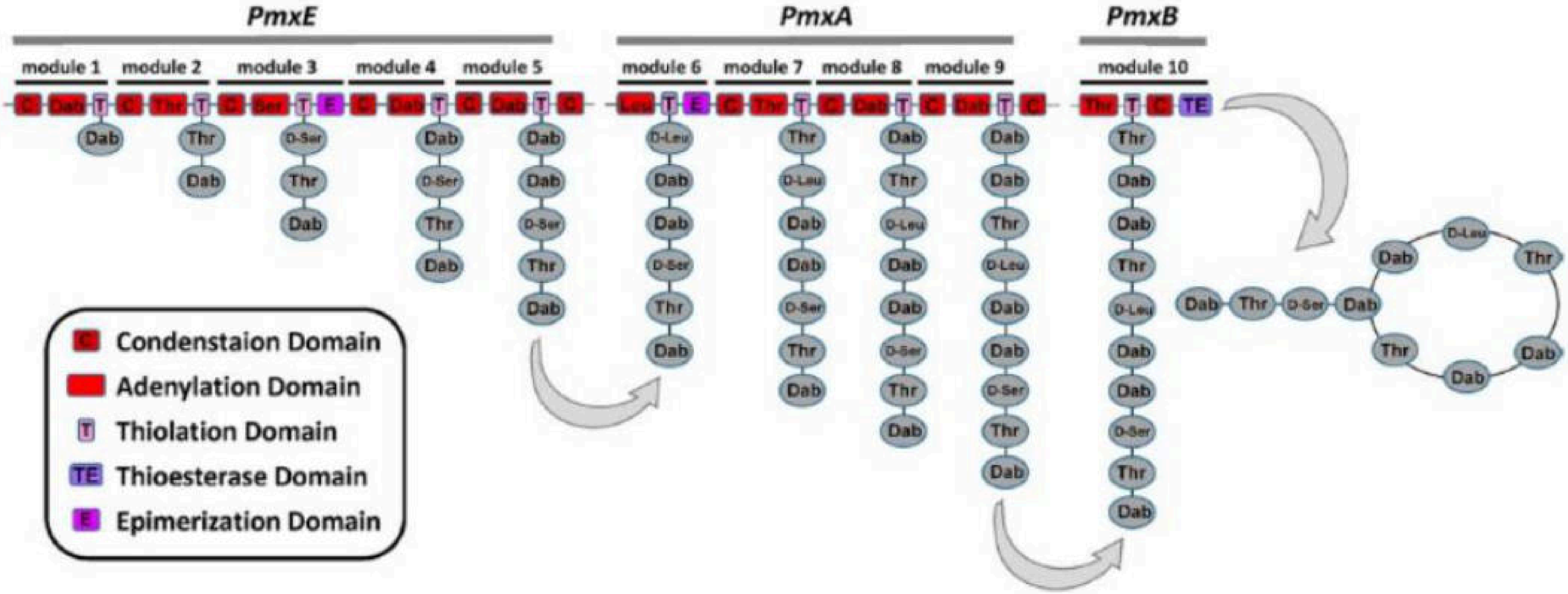
Polymyxin E (colistin)



Polymyxin B

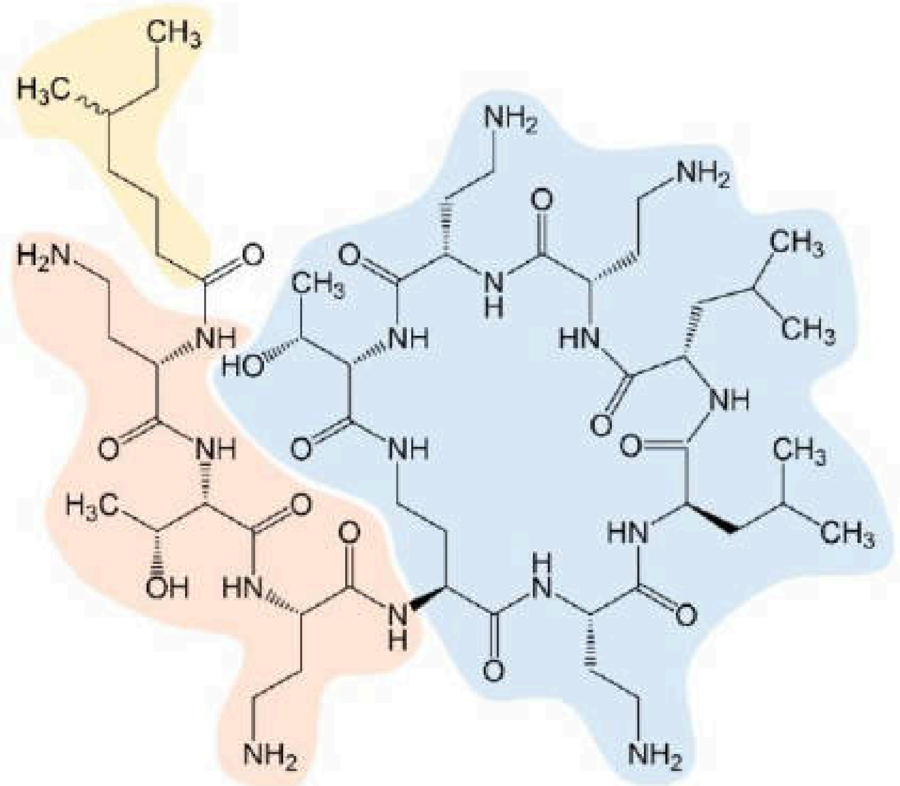
Produced by Gram positive bacteria "*Paenibacillus polymyxa*"

- Use for Gram negative bacteria infection
    - Multiple drug resistant P. aeruginosa
    - Carbapenemase-producing Enterobacteriaceae
- By breaking up bacterial cell membrane



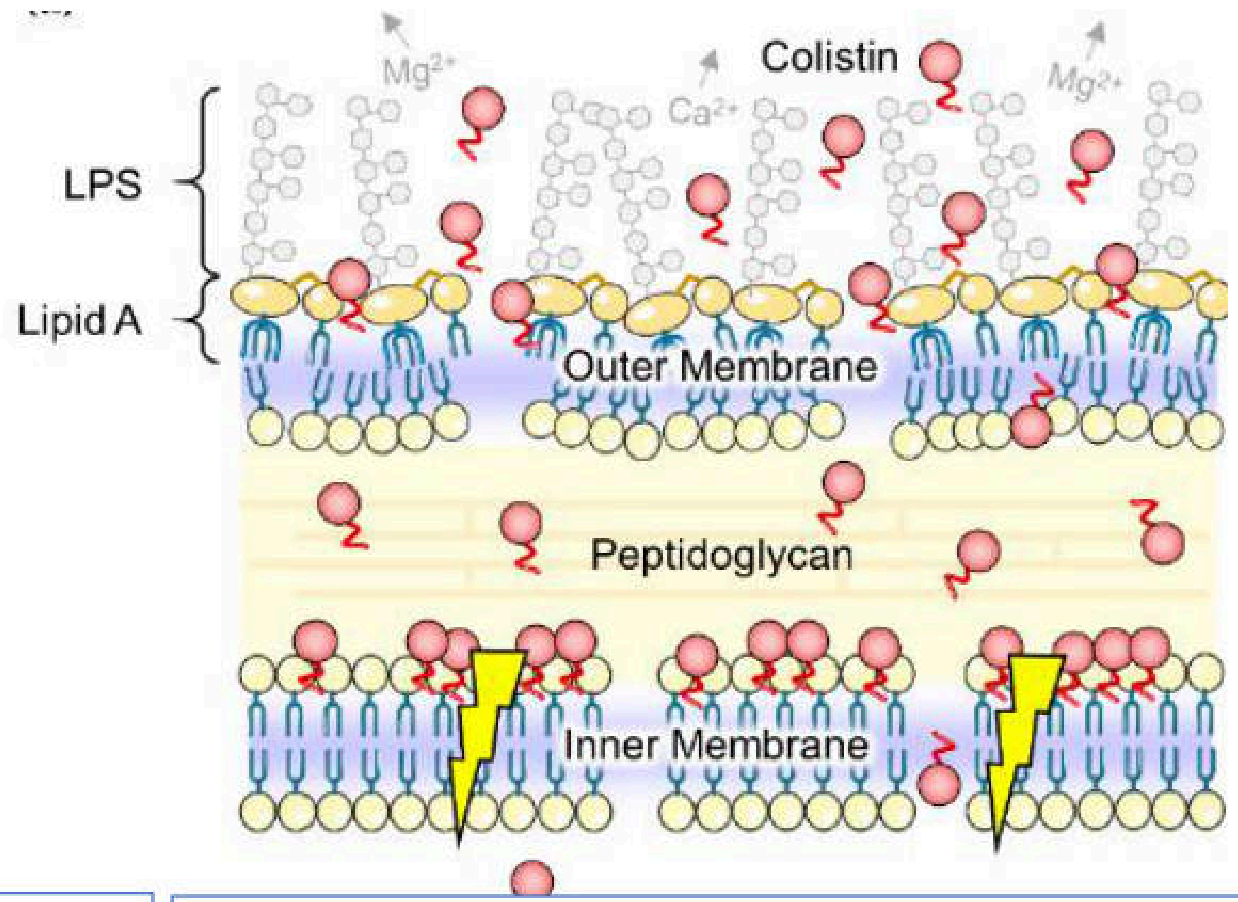
The polymyxins are produced by non-ribosomal peptide (NRP) synthetase systems in Gram-positive bacteria

https://en.wikipedia.org/wiki/Polymyxin



The chemical structure of colistin is composed of three parts:

- hydrophobic fatty-acyl tail (yellow)
- a linear tripeptide segment (orange)
- a hydrophilic, heptapeptide ring (blue)

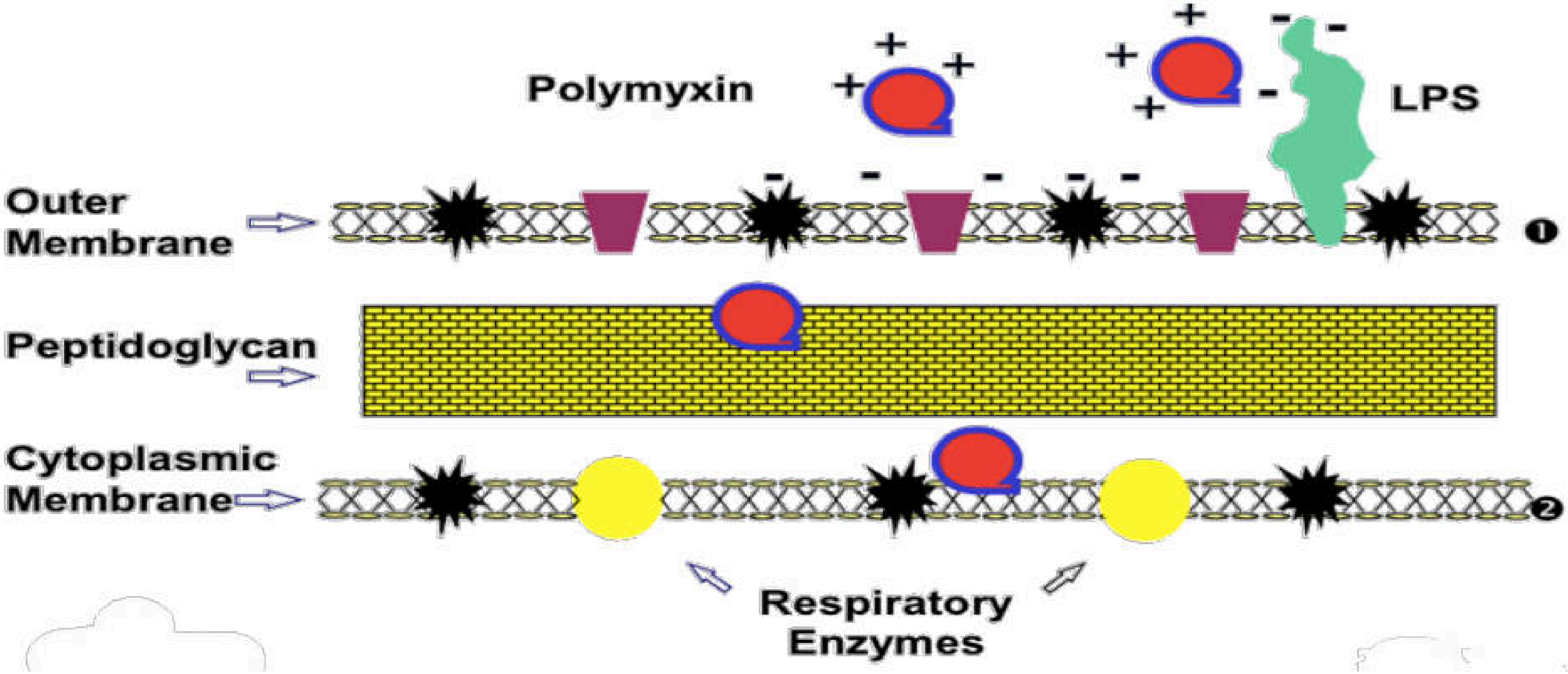


Postulated effects of colistin (red circles) on Gram- bacteria.

Colistin molecules disorganize the outer membrane and may somehow disrupt the physical integrity of the inner membrane (flash symbols) finally leading to cell death.

Peter Richter. et al. Antibiotics **2019**; 8: 158

Polymyxins: Resistance



Little or no effect on Gram-positives since the cell wall is too thick to permit access to the membrane.

Gram-positives are **naturally resistant.**

Polymyxins: Adverse Effects

- Nephrotoxicity: Increase renal tubular epithelial cell membrane permeability

  → Acute tubular necrosis, proteinuria, hematuria, azotemia, oliguria
- Neurotoxicity: block release of Ach from presynaptic

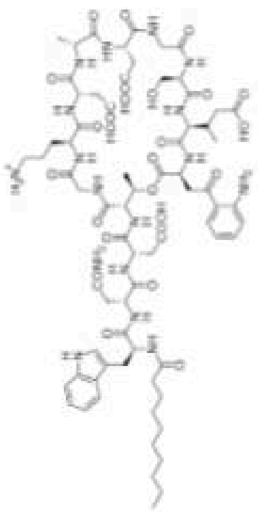
  → Facial& peripheral pareathesia, weakness, visual disturbance, ataxia, neuromascular blockade
- Respiratory arrest

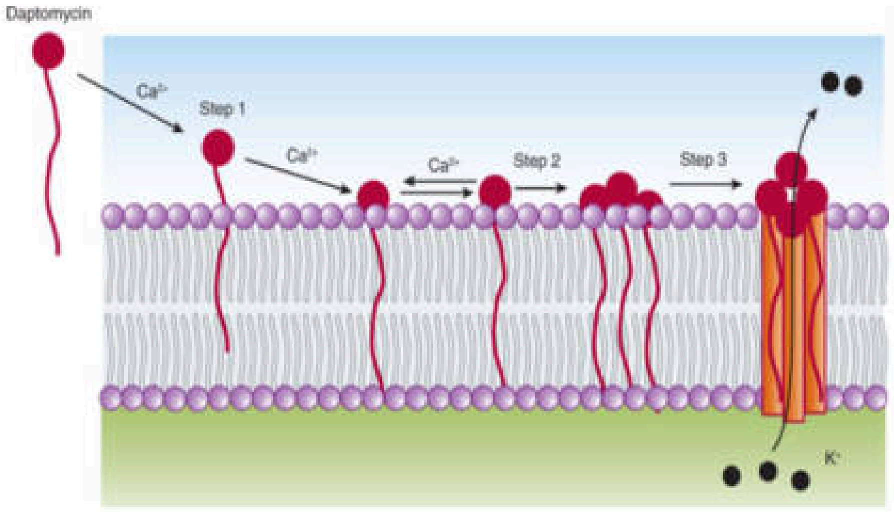
* Dose-dependent & reversible

** Avoid co-administration with nephrotoxic & neurotoxic drugs

Daptomycin: Mechanism of Action, Spectrum

- Binds to components (calcium ions) of the cell membrane
  → causes rapid depolarization with K^+ efflux
  → inhibiting intracellular synthesis of DNA, RNA, protein
- Concentration-dependent bactericidal activity





Gram (+) bacteria

- Methicillin-Susceptible & Methicillin-Resistant *S. aureus*, **Vancomycin-Resistant *S. aureus*** (**VRSA**)* and coagulase-negative staphylococci
- *Streptococcus pneumoniae* (including PRSP), Viridans streptococcus, Group streptococcus
- *Enterococcus faecium* & *faecalis*, **Vancomycin-Resistant *Enterococcus*** (**VRE**)*
- Group streptococcus

Relatively inactive against Gram (-) aerobes

Daptomycin: Pharmacokinetic

- Concentration-dependent bactericidal activity
- Only available parenterally
- Distribution: readily distributes into well-perfused tissue; protein binding = 90%
- Achieves high serum levels, **but poor penetration into CNS**, bone
- Excreted primarily by the kidneys; $t_{1/2}$ = 7.7 - 8.3 hours in normal renal function; dosage adjustments are required in the presence of RI; not removed by HD

Daptomycin: Clinical Uses

- Use reserved for serious/complicated infections caused by resistant bacteria:
  - Complicated skin and soft tissue infections due to MSSA, MRSA, or *Streptococcus pyogenes*
  - Bacteremia due to *Staphylococcus aureus*
  - Data accumulating in treatment of serious infections due to MRSA or VRE (endocarditis, meningitis, osteo); catheter-related bacteremia

- **Daptomycin** should **NOT** be used in the treatment of pneumonia (lack activity in lung parenchyma due to inactivation by surfactant)

Daptomycin: Adverse Effects

- N/V, diarrhea
- Headache
- Injection site reactions
- Rash
- Myopathy and CPK elevations
  - Interaction with **HMG CoA-reductase inhibitors**

    → increase incidence of myopathy